Mobin Asaadi

Graduate Physics Student

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Mobin7d2 mobin7d2.github.io

Research Interests

Quantum Optics, Quantum Information, Quantum Simulation

Education

Master of Physics | University of Ottawa, Ottawa, ON, Canada — May 2026 - Present

Supervisor: Professor Ebrahim Karimi

Bachelor of Physics | Amirkabir University of Technology, Tehran, Iran — Sep 2021 - August 2025

GPA: 17.94/20

Supervisor: Assistant Professor Negar Ashari Astani

Publications

Detailed Study of the Fluorescence and Inner Filter Effects of Malva sylvestreis
Journal of Photochemistry & Photobiology, A: Chemistry

DOI: 10.1016/j.jphotochem.2024.116219

Authors: Ali Bavali, Ali Rahmatpanahi, **Mobin Asaadi**, Mina shiri

Projects

Simulating Rock Candy Formation Using a Random Walk Approach

Project Description: A classical two-dimensional random walk was assumed for the movement of sugar molecules and the process was simulated using Python. This project was done during a Random Processes course at Amirkabir University of Technology.

Link: Further details on GitHub.

Diffusion Simulation

Project Description: As a final project for my Computational Physics Course at Amirkabir University of Technology, I simulated diffusion of two different types of gasses in a segmented container. Also relevant thermodynamical quantities were calculated.

Link: Further details on GitHub.

Ising Model and Curie Temperature

Project Description: In this project, a two-dimensional Ising model was simulated using Python to study the Curie temperature.

Link: Further details on GitHub.

Academic Associations, Affiliations, and Services

Co-founder and Managing editor of students' scientific journal of physics department "Meydan" — Jan 2023 - Jul 2025

As managing editor, my job was to set up meetings with the editoral board and help them improve their articles for our journal.

Secretary of Students' Scientific Association of Physics at AUT Jan 2023 - Jan2024

My team and I, selected through the student election, successfully organized several seminars and webinars addressing the needs of students in the department.

We also organized various activities for physics enthusiasts outside the university. For instance, during International Astronomy Week, we hosted a range of events and were recognized as the **3rd best team** by the Astronomical Society of Iran.

Experience

Teaching Assistant
Sep 2026 - Present

Physics Department of University of Ottawa

Course: Principles of Physics - Lab (I)
Under supervision of Prof. Michael Wong

Teaching Assistant
Sep 2024 - Aug 2025

Physics and Energy Engineering Department of Amirkabir University of Technology

Course: Quantum Mechanics (I) & (II)
Under supervision of Prof. Hossein Abbasi

Undergraduate Research Assistant
Jul 2024 - Aug 2025

Department of Physics and Energy Engineering of Amirkabir University of Technology

Under supervision of Prof. Negar Ashari Astani

Undergraduate Research Assistant
Jul 2023 - Jul 2024

Department of Physics and Energy Engineering of Amirkabir University of Technology

Under supervision of Prof. Ali Bavali

Teaching Assistant
Jan 2023 - Jun 2023

Department of Mathematics and Computer Science of Amirkabir University of Technology

Course: Calculus (II)

English as Second Language Teacher
Sep 2021 - Dec 2021

Hafez Language Institute

Awards & Certifications

GRE Subject Test in Physics: 930/990, **Percentile: 85**

National Physics Olympiad: Team member representing Amirkabir University of Technology. Ranked **20th** nationally.

National Masters Entrance Exam: Ranked **19th** among a few thousands participants.

Summer School on Mathematical Physics: Participance in summer school program focused on Mathematical Physics, organized by the Kurdish Mathematics Society. Achieved the **highest grade** (A) in the program.

3rd Best Team, National Astronomy Festival: Our team was awarded for magnificent activities during the Astronomy Week 2023 by Astronomical Society of Iran.

Summer School on Machine Learning and Applications: A comprehensive summer school program on Machine Learning and its applications, conducted by Madrese Sino.

National Olympiads for High School Students: Accepted into the second level of the National Olympiads in Physics, Mathematics, and Astronomy.

Teaching Training Course (TTC): Successfully completed a comprehensive TTC, issued by Ferdowsi University of Mashhad.

Advanced English Courses: Completed advanced courses in English language at Safir English Academy, achieving an average passing mark of **91%**.

Skills & Languages

Persian	Native
English	**IELTS Overall band: 8.0** Listening 7.5 | Reading 8 | Writing 7.5 | Speaking 8
Programming language	Python (with experience in libraries such as NumPy, Pandas, and Qiskit)
Computational software	Quantum ESPRESSO
Scientific graphing software	Origin
Text editor software	LaTeX, Microsoft Word